have the highest variance (y, in the example) and split the points such that the variance of the two new point sets is minimal; finally, we remove S_9 from its parent S_4 and add two new leaves containing the two point sets resulting from the split.
(bottom) The final result: we now need to update S_4's centroid and radius, and backtrack.

Suppose, in fact, that we have found that we should add Z to some leaf L, that already contains j points; we know that $j \geq m > 1$, so the leaf is not empty, but there could be three very different situations:

1. If L already contains Z, we just don't do anything, assuming we don't support duplicates (otherwise, we can refer to the remaining two cases).
2. $j < M$: in this case, we just add Z to the list of L's children, recompute the centroid and radius for L, and we are done. This case is shown in figure 10.12, where L==S14: on the right side of the figure you can see how the centroid and radius of the bounding envelope are update as a result of adding Z to S14.
3. $j == M$: This is the most complicated case, because if we add another point to L, it will violate the invariant requiring that a leaf holds no more than M points. The only way to solve this, is splitting the leaf's point into two sets, and creating 2 new leaves, that will be added to L's parent, N. Unfortunately, by doing this we can end up in the same situation if N already had M children: again, the only way we can cope with this is splitting N's children in two sets (defining two spheres), remove